

CS614 Final Term Current subjective shared by Students solved by Masters

2024 Feb

Mcq's 30 sy ziada Junid Wali file sy thy
Mostly mcqs from junaid file and some from handouts
Mcqs handsout but 10 from past files
Mostly from Junaid file.
MCQS mostly from Junaid file
Mostly mcqs from junaid file and some from handouts

Q. Three dimensions context of web warehouse

1. Three dimensions common in context of Web data warehouse.

Page, Event, Session, Referral

OR

Sure, here are three dimensions for a web data warehouse:

1. **Page Dimension:** This refers to the specific webpage where an event occurs. It's flexible to accommodate both static and dynamic pages.
2. **Event Dimension:** This captures what action a user takes on a webpage, like clicking a link or filling out a form.
3. **Session Dimension:** This tracks a user's entire visit to a website, from the moment they arrive until they leave.
4. **Referral:** Indicates where users came from before landing on the website, providing insights into traffic sources and marketing effectiveness.

Each dimension provides a different perspective for analyzing user behavior on a website.

Quality validation techniques

1. **Referential Integrity (RI):** Ensures that relationships between data tables are maintained, preventing orphan records.
2. **Attribute domain:** Validates that data values fall within predefined ranges or types.
3. **Data Quality Rules:** Apply specific criteria to data to check for accuracy, completeness, and consistency.
4. **Data Histogramming:** Visualizes data distributions to identify anomalies or patterns for validation purposes.

Project planning and design

1. **Project Planning:** This is the first step where you assess your organization's readiness for a data warehouse project, establish the scope, justify the costs vs benefits, and form a team.
2. **Design:** This involves understanding business requirements, setting up the technical architecture, selecting products, designing the data model, and planning for data staging and analytic applications. It's all about translating business needs into a functional design.

OR

Project planning involves assessing readiness, scoping the project, justifying costs, assembling the right team, and creating a detailed project plan with milestones and deliverables. Designing the data warehouse includes defining business requirements, creating a technical architecture, selecting

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appropriate products, dimensional modeling, physical design, and developing ETL processes to stage data effectively. These steps ensure alignment with business needs and technical capabilities for a successful data warehouse implementation.

Q. Classes of anomalies (lexical error and missing record)

Ek me lexical and missing record se

Q2. Lexical error and Missing attributes ki definition lekhni thi. (3 marks)

Classes of anomalies mn lexical error and missing records (3 marks)

1. **Lexical Error:** This is a type of syntactic anomaly where the data does not match the expected format. For example, if a table is supposed to have five columns but some rows only have four, that's a lexical error.
2. **Missing Record:** This is a type of coverage anomaly where an entire record (or row in a table) that should be present in the data is missing. This could be due to issues during data collection or processing.

OR

Lexical errors refer to mistakes in the format of data, like having fewer columns in a table row than expected. Irregularities involve inconsistencies in the use of values, units, or abbreviations, such as using different currencies to denote salaries. Missing records are instances where expected data entries are absent, leading to gaps in the dataset, requiring careful consideration to determine if they represent real-world absences or errors in data collection.

1. Three parallel track Kimball approach

1) kimball parallel tracks types. (3 marks)

2) Three tracks name Kimball approach (3 marks)

In Kimball's approach to data warehouse development, there are three parallel tracks:

1. **Technology Track:** Focuses on designing the technical architecture to support integration of various technologies. It involves evaluating and selecting specific products based on the identified capabilities.
2. **Data Track:** Translates business requirements into a dimensional model (data model), which is then transformed into a physical structure. Activities include performance tuning strategies like aggregation, indexing, partitioning, and designing ETL processes for data staging.
3. **Analytic Applications Track:** This track is about designing and developing applications that will analyze the data stored in the data warehouse.
This track focuses on creating tools and interfaces for users to analyze and derive insights from the data

2. 3 task of DT?

here are three tasks that are typically part of Data Transformation Services (DTS):

1. **Establishing Connections:** This involves setting up the necessary connections to the source and destination databases.
Setting up connections to source and destination databases.
2. **Data Import and Export:** This task involves extracting data from the source and loading it into the destination. It's like moving data from one place to another.
3. **Data Transformation:** This task is about changing the data in some way. For example, it might involve cleaning up the data, or converting it into a different format. It's like changing the data to make it more useful or easier to work with.

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Transforming data: Modifying the format or content of data during the transfer process.

3. Write names of DTS items (5 marks)

Here are the names of five items in Data Transformation Services (DTS):

1. **DTS Packages:** These are collections of connections, tasks, and workflows that are executed together.
Collections of tasks and transformations for data movement and transformation.
2. **DTS Tasks:** These are individual units of work within a package, such as establishing connections, importing/exporting data, or transforming data.
3. **DTS Transformations:** These are operations that change the data in some way during the data movement process.
Processes within tasks that modify the format or content of data.
4. **DTS Package Workflows:** These define the order in which tasks are executed within a package.
Sequences of tasks executed in a specific order to accomplish a task or process.
5. **DTS Tools:** These are utilities or applications used to create, manage, and execute DTS packages.
Software interfaces and utilities used to create, manage, and execute DTS packages and tasks.
6. **Meta Data:** Information about the structure and characteristics of data stored within a DTS package or task.

4. Two parallel Software Architecture

Parallel Software Architectures

Shared Memory	Shared everything
Shard Disk	Shared everything
Shared Nothing	

5. Two main types clustering in data mining

6. Two types of clustering in mining (3 marks)

Two main types are one way clustering and two way clustering.

One way clustering is when only data records (rows) are used. Two way clustering is when all the rows and columns are being used for clustering purpose.

OR

1. **One-way Clustering:** This method uses all the attributes of the data and clusters based on similarity among rows. It gives a global view of the data.
2. **Two-way Clustering (Biclustering):** This method clusters both rows and columns simultaneously, providing a local view of the data. It's useful for getting more detailed insights from specific clusters.

7. Cost of performing nested loop join

The cost of performing a nested loop join is calculated based on the number of blocks accessed in both tables involved in the join. It's the sum of the cost of accessing the first table and the product of the number of qualifying rows in the first table and the blocks accessed in the second table. The order of tables matters, as changing the order can significantly affect the cost. For example, if Table A has 500 blocks and Table B has 700 blocks, the join cost of A&B could be 35,500 I/Os, while the join cost of B&A could be 50,700 I/Os, an increase of about 43%.

Nested-Loop Join: Cost Formula

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the formula for estimating the join cost is:

Join cost = Cost of accessing *Table_A* +

of qualifying rows in *Table_A* × Blocks of *Table_B* to be scanned for each qualifying row.

OR

Join cost = Blocks accessed for *Table_A* +

Blocks accessed for *Table_A* × Blocks accessed for *Table_B*

OR

The cost of performing a nested loop join in a data warehouse is calculated based on the number of qualifying rows in each table and the blocks of data that need to be accessed. This cost is determined by the number of input/output (I/O) operations required to access the necessary data. The formula considers the size of each table and the number of qualifying rows in the outer and inner tables. It's essential to optimize the order of tables in the join to minimize I/O operations and overall cost.

8. Context of DWH web page dimension five attributes

Page Key

Page Source

Page Function

Page Template

Page File Name

Item Type

Graphics Type

Animation Type

Sound Type

9. DWH lifecycle key factor batane thay

10. DWH life cycle key factors.

Project Planning

Business Requirements Definition

Parallel Tracks

Deployment

Maintenance

OR

Key factor of Data Warehouse (DWH) lifecycle:

Understanding and translating business requirements into design considerations, ensuring alignment between technology, data, and analytic applications development.

OR

Key factor of DWH lifecycle:

Understanding business requirements and translating them into design considerations is a crucial factor. This involves assessing organization readiness, defining project scope, and obtaining necessary resources during project planning. Additionally, designing the technical architecture, selecting appropriate products, and developing analytic applications are vital steps in the lifecycle. Deployment and maintenance ensure the continuous functioning and improvement of the data warehouse.

11. Four categories of data quality Improvement

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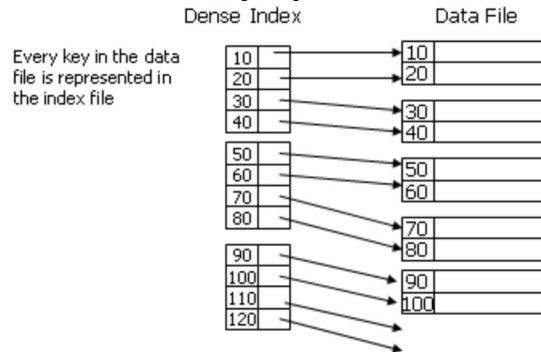
12. Four categories Data Quality improvement (5 marks)

The four categories of Data Quality Improvement

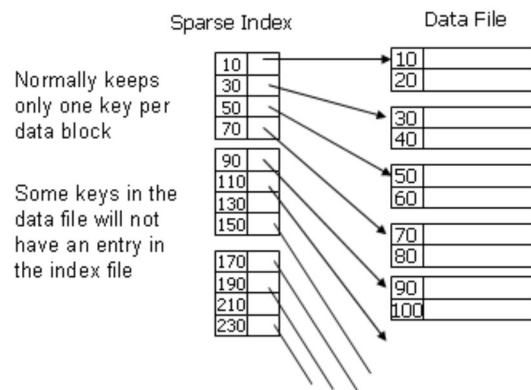
- ☐ Process
- ☐ System
- ☐ Policy & Procedure
- ☐ Data Design

1.sparse index Statement given btana correct ha ya incorrect? (3 marks)

Dense Index: Every key in the data file is represented in the index file



Sparse Index: It has some of the Record



3. Three name variants Nested-loop join? (3 marks)

1. Any three nested-join names. (3)
2. *Nested loop joining*
3. Ek Three name variants Nested-loop join?
4. 2) variant of nested loop join. (3 marks)

Nested-Loop Join: Variants

1. Naive nested-loop join
2. Index nested-loop join
3. Temporary index nested-loop join

6. Three Deliverables k names lekhny thy education Sy related tha question. (3 marks)

here are the three deliverables related to education in a data warehouse project:

1. **Data:** This is the actual information stored in the data warehouse that users need to understand.

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2. **Analytic Applications:** These are tools or software that help users analyze the data in the warehouse.
3. **Data Access Tools:** These are the tools that allow users to retrieve and interact with the data in the warehouse.

These are crucial for users to effectively use and benefit from the data warehouse.

1. Five names of Development Methodology? Just name (5 marks)

Development methodologies

- Waterfall model
- Spiral model
- RAD Model
- Structured Methodology
- Data Driven
- Goal Driven
- User Driven

2. Limitation of Using Time contiguous log entries.? (5 marks)

The method of using time-contiguous log entries to identify a session has a few limitations:

1. **Large ISPs:** Visitors from large Internet Service Providers (ISPs) may reuse the same IP addresses in a short time, making it hard to distinguish between different users.
2. **Different IP addresses:** A single user might appear to have different IP addresses within the same session, which can confuse the tracking.
3. **Firewalls:** Some firewalls can make it difficult to track a user's session accurately.

These issues make it challenging to reliably identify unique sessions using this method.

3. Shared Nothing RDBMS & partitioning k just Name? (5 marks)

1. Five type of partitioning of shared Nothing RDBMS (5 marks)?

Shared Nothing RDBMS Partitioning:

1. Hash partitioning
2. Key range partitioning
3. List partitioning
4. Round-Robin partitioning
5. Combinations of partitioning methods (e.g., Range-Hash, Range-List)

5. First two steps in BSN method. (3)

First two steps in the Basic Sorted Neighborhood (BSN) method:

1. **Create Keys:** Compute a unique identifier (key) for each record by extracting relevant fields or portions of fields.
2. **Sort Data:** Arrange the records in the sequential list according to the computed keys.

6) BSN two methods (3 marks)

Sliding Window

Selection of Keys

Problem with keys

Matching Candidates

Equational Theory

Rule Language

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Mismatching Candidates Distance Metrics

6. Any three parallel hardware architectures. (3)

Parallel Hardware Architectures

- ☐ Symmetric Multi Processing (SMP)
- ☐ Distributed Memory or Massively Parallel Processing (MPP)
- ☐ Non-uniform Memory Access (NUMA)

7. A diagram of Processes A, B, C, D was given. Business impact on y-axis against Feasibility on x-axis. Which process would have the highest priority? (3)

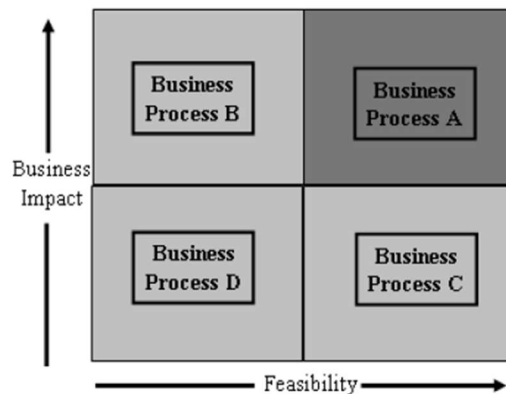


Figure -33.2: The quadrant method

So the process having higher feasibility and impact is given higher priority over the process having lower feasibility and impact. In example of Figure 33.2, process A has highest priority while the process D has lowest priority.

8. In planning phase of DWH Lifecycle, define the key factors for the readiness of a project. (5)

Here are the key factors for the readiness of a project in the planning phase of the Data Warehouse (DWH) Lifecycle:

1. **Business Sponsor:** This is the person who has a clear vision for the data warehouse and can convince others in the organization to support the project.
2. **Business Motivation:** There should be a strong, compelling business reason for building the data warehouse. It should solve critical business problems.
3. **Feasibility:** This refers to whether it's practical to build the data warehouse. It includes considerations like whether the necessary data is available and whether the organization has the technical resources to build and maintain the data warehouse.
4. **Business/IT Relationship:** A strong relationship between the business side and the IT side of the organization is important for the success of the project.
5. **Culture:** The organization's culture should be supportive of the data warehouse project.
6. **Scoping:** This involves defining what the data warehouse project will and will not include. It's important to have a clear scope to ensure the project stays on track.

9. List some reasons for why Web Warehousing may be important. (5)

here are some reasons why Web Warehousing may be important:

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1. **Web Mining:** Web warehousing can help mine the vast content of the web to find specific information, much like finding a needle in a haystack.
To extract valuable information from the vast amount of unstructured and dynamic web data.
2. **Analyzing Web Traffic:** Web warehousing can be used to analyze web traffic, which can be useful for website owners, e-commerce businesses, and advertisers.
Understanding and measuring user interactions on websites for various purposes such as improving user experience, evaluating marketing campaigns, and optimizing website performance.
3. **Archiving the Web:** The web is dynamic and constantly changing. Web warehousing allows us to archive this content for future use.
Preserving the dynamic nature of web content for future reference, historical analysis, and various projects, including research, e-commerce, and cultural preservation.
4. **E-Commerce:** In our data-driven economy, commerce is increasingly happening online. A web warehouse can serve as the equivalent of a commercial data warehouse in this context, facilitating transactions beyond geographical boundaries.

10. Related to Quality Measurement Grid. A table was given. Write the name of the stage for the given situations. (5)

Q. Quality management maturity grid table tha us mai stages kay name likhne thay

Quality Management Maturity Grid

	Management understanding	Quality organ. status	Problem handling	Cost of quality % of sales	Company attitude
Stage-1 Uncertainty	No comprehension of quality.	Quality Dept. part of manufacturing or engineering.	Fire fighting approach.	Reported unknown, actual high.	No organized activity.
Stage-2 Awakening	Recognize quality management may be of value.	Quality Dept. still part of manufacturing or engineering.	Short term solutions, no long term approach.	Reported as low, actually high.	All talk no real action.
Stage-3 Enlightenment	Become supportive and helpful.	Quality Dept. reports to top management.	Problems faced and solved orderly.	Reported as medium actually on the higher side.	Identifying and resolving problems
Stage-4 Wisdom	Understand absolutes of quality management.	Senior quality manager position.	Identified at an early stage.	Reported as about medium actually about medium.	Defect prevention is a routine.
Stage-5 Certainty	Quality management essential part of company policy.	Quality manager on board of directors.	Identified and resolved at an early stage.	Reported as low, actually is low.	Know why there are problems.

Table-23.1: Quality Management Maturity Grid

2. Quality Management Grid keys?

- Uncertainly
- Awaking
- Enlightenment
- Wisdom
- Certainly

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Ek hash join method se long aya tha

1. Two tables are given A and B. A have 1000 columns and B have 2000 column

Calculate the cost of hash join? (5 marks)

Cost of Hash-Join

- In partitioning phase, read + write both operations requires $2(M+N)$ I/Os.
- In matching phase, read both requires $M+N$ I/Os.

4) values given which is best SQL query or manual checking (3 marks)

Number of records corrupted

If the number of corrupted records are very large in number then we need to update it through SQL Queries or some other automated way, otherwise, if they are less in number then they can be updated through inspection or manual checking.

OR

let's consider a scenario where we have a database of students with thousands of records.

SQL Query Advantage:

Suppose you want to find all students who scored above 90 in a particular subject. Using an SQL query would be the best approach here. The query might look something like this:

```
SELECT * FROM students WHERE score > 90;
```

This would instantly give you a list of all students who scored above 90. Doing this manually by looking through each record would be time-consuming and prone to errors.

Manual Checking Advantage:

Now, suppose you have a small set of data, say 10 records, and you want to review each record to understand the data better or to find some qualitative aspect that's not easily quantifiable. In this case, manual checking might be more suitable.

For example, if you want to understand the writing style of each student from a small set of essay submissions, you would need to manually read through each essay as this cannot be effectively done using an SQL query.

2) Find cost of hash in matching phase. (5 marks)

The cost of the matching phase in a hash join operation is related to the number of input/output operations (I/Os) required to read the data.

the cost of the matching phase in a hash join operation is $M + N$ I/Os.

3) DWH lifecycle phases (5 marks)

1. Project Planning
2. Business Requirements Definition
3. Parallel Tracks
4. Deployment
5. Maintenance

4) sources of web scraping also explain three reasons (5 marks)

During the last decade the web has grown from a curiosity to a required information repository. From news headlines and stories, financial data and reports, entertainment

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events, online shopping and business to business commerce, the Web has it all. The promise of the web is that all of this information will be easily maintained, monitored, queried, and browsed.

Some of the uses of scrapping are:

- Building contact lists
- Extracting product catalogs
- Aggregating real-estate info
- Automating search Ad listings

2022

15 mcqs past + 25 mcqs handout
75% of quizzes from sir Junaid file.

Sql queries long

Q3. Table data was given we had to count incorrect and correct data using SQL queries (business rules validate ky related tha)

Ans:

```
SELECT  
SUM(CASE WHEN total_amount > 100 AND shipping_address IS NOT NULL THEN 1 ELSE 0 END) AS  
correct_orders,  
SUM(CASE WHEN total_amount <= 100 OR shipping_address IS NULL THEN 1 ELSE 0 END) AS  
incorrect_orders FROM Orders;
```

3) records of student were given. 2 Sql query likhni thi. Aik query main data of birth wali field main null entries ki jaga given date likhni thi or Dosri query main distinct student names lany thay ascending order main.

Ans:

```
UPDATE students  
SET dob = 'given date'  
WHERE dob IS NULL;  
SELECT DISTINCT student_name  
FROM students  
ORDER BY student_name ASC;
```

Q6 differentiate between data mining and statistics on basic of data type and dimensions

Data mining and statistics differ based on data type and dimensions. Statistics typically deals with smaller datasets and simpler relationships, often linear, while data mining handles larger datasets with complex, nonlinear relationships among many variables.

3. Diffrenet metrics used for comparison data mining techniques. Name the metrics?

- Accuracy

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- Speed
- Robustness
- Scalability
- Interpretability

4. Time complexity of K means $O(tkn)$, $t=?$ $k=?$ $n=?$

$O(tkn)$, where n is # objects, k is # clusters, and t is # iterations. Normally, $k, t \ll n$.

OR

In the time complexity formula $O(tkn)$ for K-means clustering:

- " t " represents the number of iterations (how many times the algorithm repeats)
- " k " represents the number of clusters (groups) you want to classify your data into. You decide this number when you start the algorithm.
- " n " represents the number of objects (data points) in the dataset.

5. Identity header row and how columns delimited?

the first row contains the header and the columns are delimited by comma

6. Situation when Parallelism can be exploited.

Parallelism can be exploited in the following situations:

1. **Symmetric Multi-Processors (SMP), Clusters, or Massively Parallel (MPP) Systems:** These are systems with multiple processors that can run processes simultaneously.
2. **Sufficient I/O Bandwidth:** There should be enough bandwidth to handle the data transfer needs of the parallel processes.
3. **Underutilized or Intermittently Used CPUs:** If the CPU usage is typically less than 30%, it means there's enough processing power available for parallel processes.
4. **Sufficient Memory:** There should be enough memory to support additional memory-intensive processes such as sorts, hashing, and I/O buffers.

7. Diagram illustrating Business dimensional life cycle.

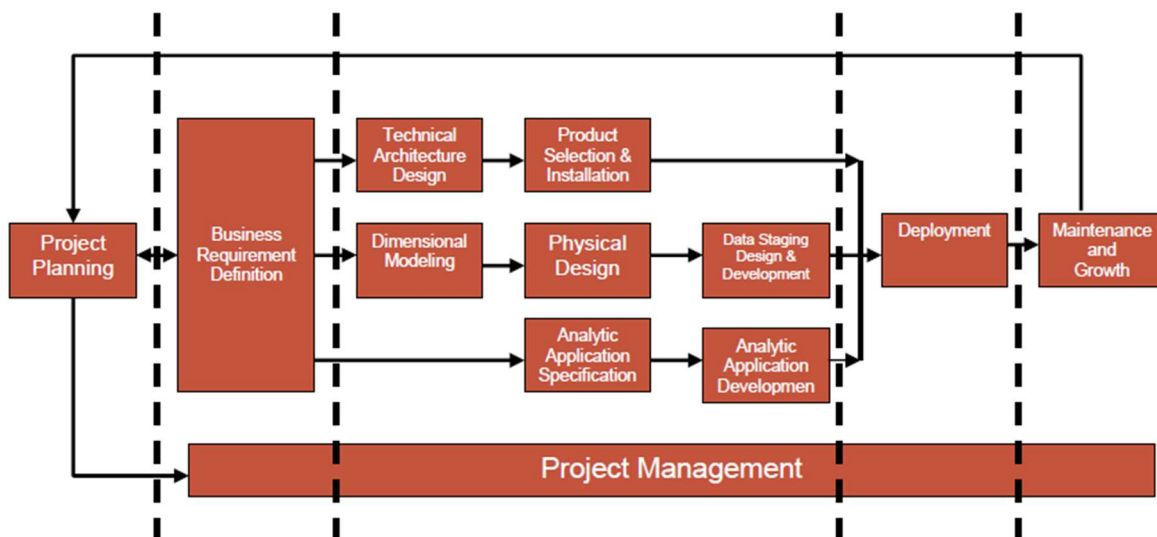


Figure -33.1: Business Dimensional Lifecycle (Kimball's Approach)

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34. Types of Parallelism

ANSWER:

There are two types

- Data Parallelism
- Spatial Parallelism

31. Issues of Click stream.

ANSWER:

Clickstream data has many issues

- Identifying the Visitor Origin
- Identifying the Session
- Identifying the Visitor
- Proxy Servers
- Browser Caches

EXTRA

Q4. 2 Barriers to Linear Speedup

Ans:

Amdahl' Law

Startup

Interference

Skew

3. Static and dynamic Attributes

Static Attributes	Dynamic Attributes
Farmer Name	Date of Visit
Farmer Address	Pest Population
Field Acreage	CLCV
Variety(ies) Sown	Predator Population
Sowing date	Pesticide Spray Dates
Sowing method	Pesticide(s) Used

Classes of Anomalies

- Syntactically Dirty Data
 - Lexical Errors
- Semantically Dirty Data

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- Duplication
- Coverage Anomalies
- Missing Records

2) data mining techniques accuracy, speed, robustness, scalability, interpretability.

2 statements were given. In main say batana tha kon si hain

Ans:

Accuracy and Speed:

If one statement emphasizes the importance of achieving the highest possible accuracy in data mining results, it likely prioritizes accuracy over speed.

Robustness and Scalability:

If one statement discusses the need for data mining techniques that can handle noisy or imperfect data and adapt to different situations, it focuses on robustness.

Interpretability:

If one statement stresses the importance of easily understandable and interpretable results from data mining models, it values interpretability.

Write any two implementation strategies of data warehouse project (3 marks)

1) names of data implementation strategies

There are two Strategies of Data implementation:

- Top-Down Approach
- Bottom-Up Approach

Top-Down Approach:

Top-Down Approach is generally useful for projects where the Technology is mature and well understand.

Bottom-Up Approach:

This approach is used when the Business goal is achieved by Data Warehouse.

What are the load tasks using import/export wizard. Write any 3 (3 marks)

First of all load data for the Lahore campus

1. Connect to source Text files
2. Connect to Destination SQL Server
3. Create new database 'Lahore_Campus'
4. Create two tables Student & Registration
5. Load data from the text files containing student information into Student table
6. Load data from the text files containing registration records into Registration Table

Name three major approaches for accessing information stored on the web. (3 marks)

1. How to access the data on the web warehouse name three approaches

Three major types of searches, as follows:

1. Keyword-based search
2. Querying deep Web sources
3. Random surfing

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